Table 2 (null effect, null_pilot, $B = 48$ per cell). Estimated exposure effects of cumulative total exposure when the true effect is zero, so both risk models are correctly specified (bold cells with correct model specification).

	Estimated β (true: 0 for both models)								Pairs
	True x		Naive		RRC corrected				dropped
$\rho(c, C)$	Cox	ERR	Cox	ERR	Cox.A	Cox.M	ERR.A	ERR.M	(%)
A. Mean $\hat{\beta}$ over converged fits									
Additive	Cox and ERR true $\beta = 0$, additive independent error								
0.3	−0.02	0.06	−0.05	0.04	**−0.06**	−0.11	**0.08**	0.09	
0.6	0.02	0.15	0.01	0.12	**0.01**	−0.01	**0.19**	0.35	
0.9	0.05	0.31	0.03	0.21	**0.05**	0.04	**0.31**	0.33	
Multiplicative	Cox and ERR true $\beta = 0$, multiplicative independent error								
0.3	−0.03	0.05	−0.03	0.02	−0.04	**−0.05**	0.03	**0.03**	
0.6	−0.07	0.05	−0.05	0.03	−0.06	**−0.07**	0.05	**0.04**	
0.9	−0.06	0.05	−0.05	0.04	−0.07	**−0.07**	0.06	**0.06**	
B. Median $\hat{\beta}$ over all fits; [number of fits not converged or stopped at the cap $\beta = 50$]									
Additive	Cox and ERR true $\beta = 0$, additive independent error								
0.3	−0.06	−0.06	−0.06	−0.06	**−0.05**	−0.09	**−0.05**	−0.08 [3]	28.4
0.6	0.12	0.13	0.06	0.07 [1]	**0.10**	0.11	**0.12 [1]**	0.13	12.4
0.9	0.13	0.15 [1]	0.10	0.11 [1]	**0.14**	0.12	**0.15 [1]**	0.14 [1]	4.3
Multiplicative	Cox and ERR true $\beta = 0$, multiplicative independent error								
0.3	−0.05	−0.04 [6]	−0.04	−0.05 [5]	−0.04	**−0.07**	−0.06 [5]	**−0.07 [5]**	0.0
0.6	−0.04	−0.04 [7]	−0.04	−0.05 [7]	−0.07	**−0.07**	−0.07 [7]	**−0.07 [6]**	0.0
0.9	−0.02	−0.02 [7]	−0.03	−0.02 [9]	−0.04	**−0.04**	−0.03 [9]	**−0.03 [9]**	0.0

Cox model $\lambda(t) = \lambda_0(t)\exp(\beta x(t))$ and ERR model $\lambda(t) = \lambda_0(t)(1 + \beta x(t))$, both with true $\beta = 0$: the hazard does not depend on x, so Cox and ERR truth generate the same data and every fit should estimate 0. True x: the same risk model fitted to the true cumulative exposure.

$n_1 = 2000$, $n_2 = 500$ (at most 8 pairs each), $T^* = 60$, cumulative incidence about 10%. Surrogate C has compound symmetry ($\rho = 0.5$) across years, $E(C) = 0.012$; true $c = \gamma_1 C + \epsilon$ with $\gamma_1 = 2/3$ (additive) or $\log c = \gamma_0 + \log C + \epsilon$ with $E(c) = 0.008$ (multiplicative), σ_ϵ set by $\rho(c, C)$; median x 0.21 (additive) and 0.17 (multiplicative).

Pairs dropped: share of validation pairs with $c \leq 0$ or $C \leq 0$, which the log (.M) calibration cannot use. – : no fit converged, or not recorded. A median of 50.00 means at least half of the fits stopped at the cap $\beta = 50$.